EPM213s - Electrical Measurements

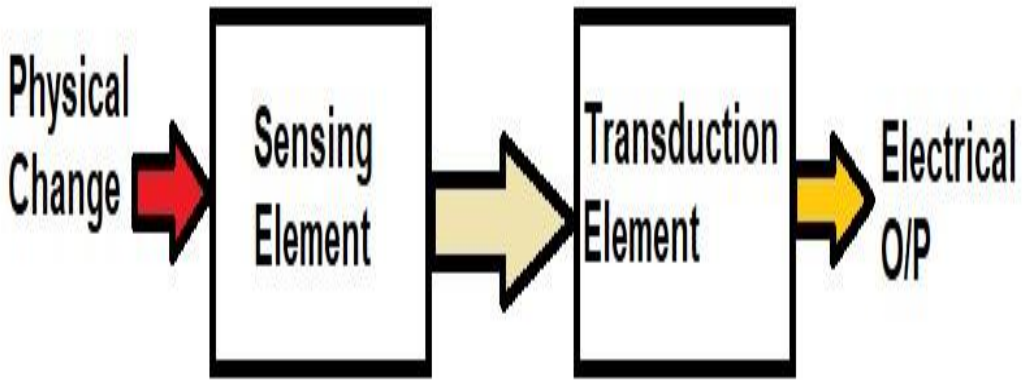
Exp 2 Transducers

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Spring 25



Transducers: Photo Resistor Experiment

1. objective

The objective of this experiment is to examine the behavior of a photoresistor (LDR - Light Dependent Resistor) by investigating the relationship between the current (I) and the distance (d) of a light source. This experiment seeks to illustrate how the resistance of a photoresistor varies with changing light intensities, influencing the measured current.

2. procedure

1. **Assemble the circuit:** Connect the photoresistor in series with an appropriate power source and a resistor to measure the current.

2. **Position the light source:** Place the light source at varying distances (d) from the photoresistor.

3. **Measure the current:** Use an ammeter to record the current (I) at different distances from the light source.

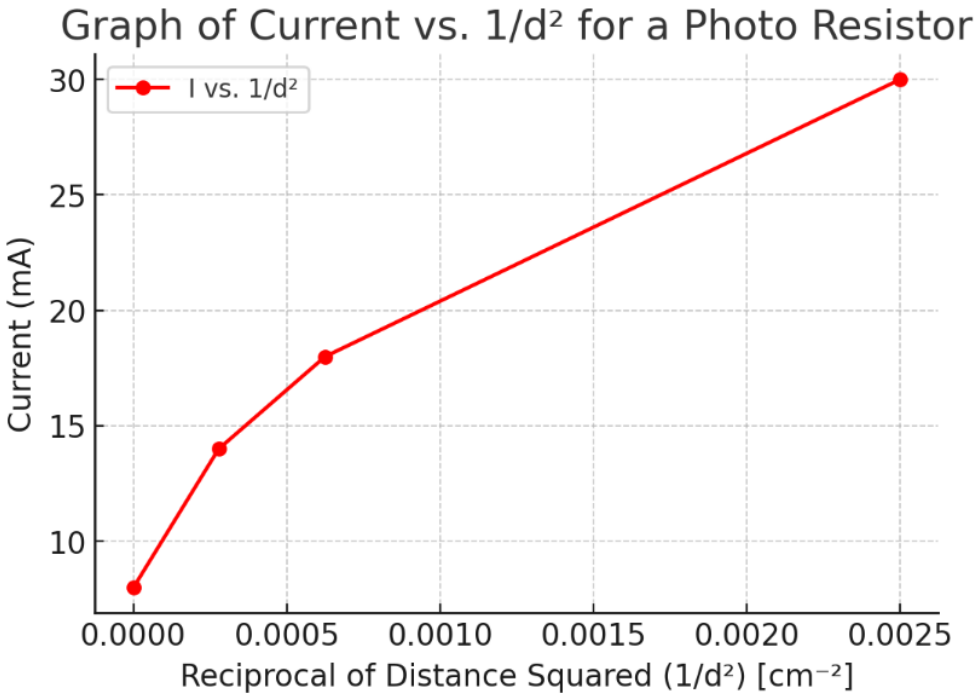
4. **Document the data:** Record the current readings corresponding to each distance.

5. **Plot the graph:** Graph the relationship between distance (d) and current (I) to analyze the dependency.**3. Equipment**

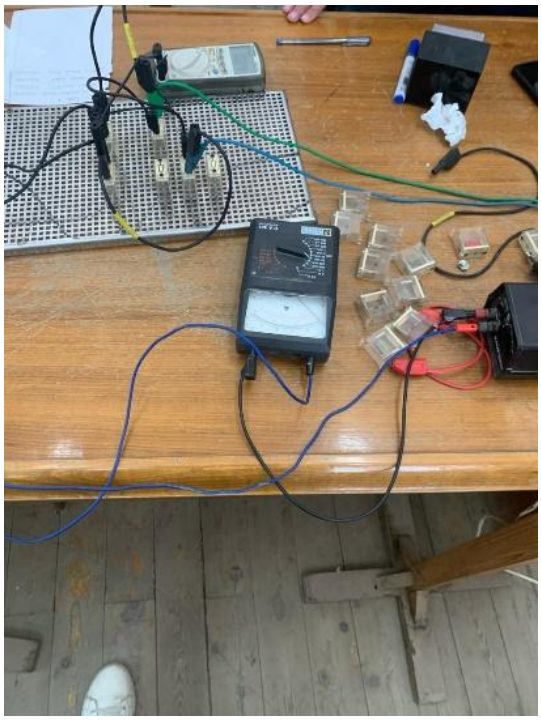
Observations and Data Table

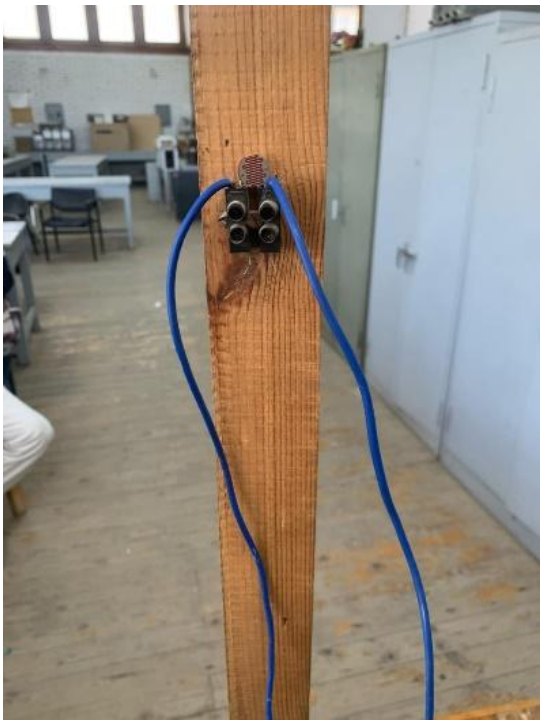
Distance (d)	Current (I)
0 cm	8 mA
60 cm	14 mA
40 cm	18 mA
20 cm	30 mA

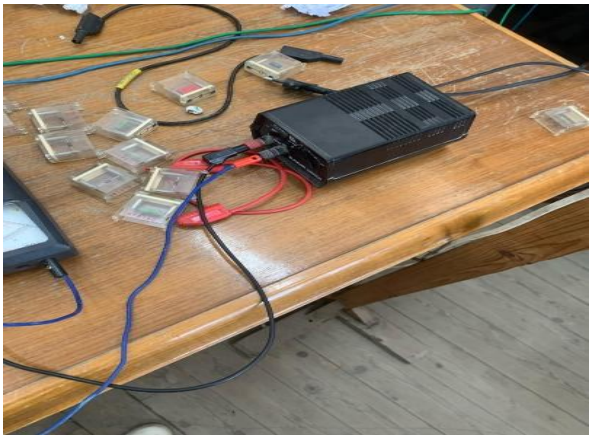
4. Graph: Current (I) vs. Reciprocal of Distance Squared ($1/d^2$)



5. Photos







Conclusion

From the experiment, we observe that as the distance (d) between the light source and the photoresistor increases, the current (I) also increases. This indicates that the resistance of the photoresistor decreases with greater light exposure, permitting more current to flow through the circuit.